

# Answer Key & Teacher Notes

Session 1 – Writing Algebraic Expressions • Session 2 – Evaluating Algebraic Expressions

## HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

|                                  |                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Warm-Up</b>                   | First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.                                                                                                                             |
| <b>Prerequisite Practice A/B</b> | Before the lesson, for students who need the skill the session uses. Key in that folder.                                                                                           |
| <b>Guided Notes</b>              | During the lesson. Students fill in the blanks as you teach; answers below.                                                                                                        |
| <b>Practice Form A</b>           | In class, after the lesson. Two pages. The self-check code tells students if they are on track.                                                                                    |
| <b>Practice Form B</b>           | Same skills, new numbers: reteach, retake, small group or absent students.                                                                                                         |
| <b>Exit Ticket</b>               | Last 5 minutes. Two per page; cut in half.                                                                                                                                         |
| <b>Homework Packet</b>           | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 • TEACHER NOTES

Writing Algebraic Expressions

### TALK PROMPTS

- How do you decide which quantity in a story should become the variable?
- The \$45 car-wash fee shows up only once. How does the expression show that it does not repeat?

### PARTNER TASK

Concert Tickets and Two Items. One partner writes  $8t + 25$  for \$8 tickets and a \$25 fee and says what  $8t$  means; the other writes  $2j + 3f$  for juice at \$2 a cup and muffins at \$3 each. Switch and check each other's variable definitions.

### WATCH FOR

- Putting the variable with the one-time fee ( $11 + 6b$  instead of  $11b + 6$ ).
- Adding instead of multiplying for a per-item rate ( $5 + d$  instead of  $5d$ ).
- Reversing "less than" ( $4 - y$  for "4 less than  $y$ ").

## SESSION 2 • TEACHER NOTES

Evaluating Algebraic Expressions

### TALK PROMPTS

- Why must you substitute the number before you multiply or add? Explain the order.
- In  $12g + 20$ , the gigabytes doubled from 3 to 6, but the cost did not. Why does the 20 keep that from happening?

### PARTNER TASK

Team Points and Words to Symbols. One partner evaluates  $3w + 2t$  for 5 wins and 4 ties (23); the other for 8 wins and 0 ties (24). Switch roles to write "8 less than  $n$ " ( $n - 8$ ) and " $n$  divided by 4, then increased by 3" ( $n \div 4 + 3$ ).

### WATCH FOR

- Writing digits side by side after substituting ( $4x$  with  $x = 3$  read as 43).
- Adding before multiplying ( $4 + 2p$  with  $p = 5$  as  $6 \times 5 = 30$ ).
- Multiplying the coefficient before squaring ( $5k^2$  with  $k = 3$  as  $15^2$ ).

## ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

| Sheet              | Standard  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|--------------------|-----------|---|---|---|---|---|---|---|---|---|----|----|
| <b>S1 • Form A</b> | 6.EE.A.2a | 1 | 2 | 1 | 2 | 1 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S1 • Form B</b> | 6.EE.A.2a | 1 | 2 | 1 | 2 | 1 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form A</b> | 6.EE.A.2c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form B</b> | 6.EE.A.2c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

## GUIDED NOTES

**variable:** A **letter** that stands for an unknown number.

**coefficient:** The number **multiplied** by the variable.

**constant:** A number by itself that does not **change**.

**algebraic expression:** An expression with at least one **variable**.

- An amount that repeats “each” time goes with the **variable**.
- A one-time amount is the **constant**.
- Order matters: “4 less than  $x$ ” is  $x - 4$ .

**Example:** The \$9 repeats each hour:  $9 \times h = 9h$ . The one-time fee is the constant: **3**. Join the terms with a plus sign. Answer: The cost is  **$9h + 3$**  dollars.

## YOU TRY

1.  $n + 6$  b.  $4k$
2.  $10t + 4$

## EXIT TICKET

1.  $y - 9$  b.  $7n$
2.  $15m + 20$
3. **A** (8)

## PRACTICE FORM A

1.  $n + 7$  b.  $5x$  c.  $y - 4$  d.  $w \div 3$  e.  $2t + 1$  f.  $3k - 6$
2. a.  $6m$ ; coefficient 6 b. 12; it is paid once, so it does not change with  $m$  c.  $6m + 12$
3.  $8t + 25$
4.  $7c - 45$
5.  $17m + 40$
6. Rani is wrong. \$5 is paid for each day, so multiply:  $5d$ . ( $5 + d$  adds only one \$5.)
7.  $3c + 2k$
8. a.  $25m + 40$  b.  $30m$  c. Gym A:  $25 \times 6 + 40 = \$190$ ; Gym B:  $30 \times 6 = \$180$ . Gym B costs less for 6 months.
9. Bela put the variable with the one-time shipping fee. The \$11 repeats for each book, so it goes with  $b$ :  $11b + 6$ .
10. **A** ( $n + 7$ )
11. **B** ( $p$ )

**Self-check word** SYMBOL

## PRACTICE FORM B

1. a.  $p + 9$  b.  $8z$  c.  $r - 10$  d.  $b \div 6$  e.  $3s + 4$  f.  $5j - 2$
2. a.  $9m$ ; coefficient 9 b. 16; it is paid once, so it does not change with  $m$  c.  $9m + 16$
3.  $11t + 20$
4.  $6c - 35$
5.  $12m + 25$
6. Tariq is wrong. \$4 is paid for each hour, so multiply:  $4h$ . ( $4 + h$  adds only one \$4.)
7.  $2j + 3f$
8. a.  $20m + 30$  b.  $24m$  c. Gym A:  $20 \times 5 + 30 = \$130$ ; Gym B:  $24 \times 5 = \$120$ . Gym B costs less for 5 months.
9. Omar put the variable with the one-time shipping fee. The \$9 repeats for each book, so it goes with  $k$ :  $9k + 5$ .
10. **C** ( $5x$ )
11. **B** ( $y - 4$ )

**Self-check word** LETTER

## GUIDED NOTES

**substitute:** Replace a variable with a given number.

**evaluate:** Find the **value** after you substitute.

**variable:** A letter that stands for a **number**.

- To evaluate, **substitute** the value for the variable.
- Then use the **order** of operations: powers, then  $\times$  and  $\div$ , then  $+$  and  $-$ .
- $4x$  means  $4 \times x$ . When  $x = 3$ ,  $4x = 12$ , not 43.

**Example:** Substitute:  $7 \times 4 - 2$ . Multiply first:  $7 \times 4 = 28$ . Subtract:  $28 - 2 = 26$ . Answer:  $7p - 2 = 26$  when  $p = 4$

## YOU TRY

- a. 16 b. 5
- $3 \times 5 + 6 = 15 + 6 = \$21$

## EXIT TICKET

- a. 17 b. 26
- $30 - 4 \times 6 = 30 - 24 = 6$  L
- A** (14)

## PRACTICE FORM A

- a. 33 b. 35 c. 25 d. 22 e. 40 f. 20
- a.  $4 \times 3 + 5 = \$17$  b.  $4 \times 8 + 5 = \$37$  c.  $\$37 - \$17 = \$20$  d. The 5 is a flat fee added once to every cost, no matter how many hours.
- $9 \times 4 + 14 = 36 + 14 = \$50$
- $5 \times 3^2 - 10 = 5 \times 9 - 10 = 45 - 10 = 35$
- Yara added  $4 + 2$  first.  $2p$  means  $2 \times p$ , so multiply first:  $4 + 2 \times 5 = 4 + 10 = 14$ .
- a.  $25 - 3 \times 2 = 19$  L b.  $25 - 3 \times 8 = 1$  L c. As d grows, 3d grows, so less water is left.
- a.  $3 \times 5 + 2 \times 4 = 15 + 8 = 23$  points b.  $3 \times 8 + 2 \times 0 = 24$  points
- a.  $12 \times 3 + 20 = \$56$  b.  $12 \times 6 + 20 = \$92$  c. Only the 12g part doubles (36 to 72). The \$20 fee stays the same, so the total does not double.
- "Less than" flips the order: start with  $x$ , then subtract 15. Correct:  $x - 15$ ; when  $x = 22$ ,  $22 - 15 = 7$ .
- A** (15)
- A** (number)

**Self-check word** EVALUATE

## PRACTICE FORM B

- a. 24 b. 27 c. 20 d. 54 e. 8 f. 18
- a.  $6 \times 2 + 7 = \$19$  b.  $6 \times 5 + 7 = \$37$  c.  $\$37 - \$19 = \$18$  d. The 7 is a flat supply fee added once to every cost, no matter how many hours.
- $8 \times 6 + 15 = 48 + 15 = \$63$
- $3 \times 4^2 + 7 = 3 \times 16 + 7 = 48 + 7 = 55$
- Ken added  $6 + 3$  first.  $3p$  means  $3 \times p$ , so multiply first:  $6 + 3 \times 4 = 6 + 12 = 18$ .
- a.  $12 + 5 \times 3 = 27$  L b.  $12 + 5 \times 9 = 57$  L c. As h grows, 5h grows, so there is more water.
- a.  $3 \times 6 + 2 \times 2 = 18 + 4 = 22$  points b.  $3 \times 7 + 2 \times 0 = 21$  points
- a.  $12 \times 4 + 20 = \$68$  b.  $12 \times 8 + 20 = \$116$  c. Only the 12g part doubles (48 to 96). The \$20 fee stays the same, so the total does not double.
- "Less than" flips the order: start with  $y$ , then subtract 12. Correct:  $y - 12$ ; when  $y = 30$ ,  $30 - 12 = 18$ .
- C** (30)
- B** (5)

**Self-check word** VARIABLE

## HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

## SHEET 1 • WRITING ALGEBRAIC EXPRESSIONS

- a.  $n + 9$  b.  $y - 8$  c.  $6k$  d.  $m \div 5$  (or  $m/5$ ) e.  $2x + 3$  f.  $7(a + 2)$
- $15L + 20$
- a.  $p \div 4$  b.  $50 - s$
- $11p + 5$ ; when  $p = 4$ :  $11 \times 4 + 5 = \$49$
- He put the variable with the one-time fee. The \$7 repeats for each ticket, so it goes with  $n$ :  $7n + 3$ .

## SHEET 2 • EVALUATING ALGEBRAIC EXPRESSIONS

- a. 30 b. 23 c. 22 d. 40 e. 12 f. 10
- a.  $3 \times 4 + 2 \times 5 = 22$  b.  $16 - 5 = 11$
- \$13, \$29, \$53
- $100 - 6 \times 5 = 100 - 30 = 70\%$
- 8 (multiply first:  $3 \times 4 = 12$ , then  $20 - 12 = 8$ )
- C** (8)
- D** ( $9t + 6$ )